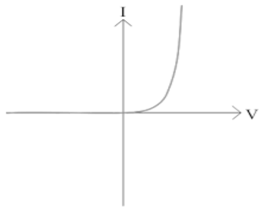
Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1.	(a)			Variable resistor added in series with lamp (1) accept any size box with an arrow through it Voltmeter added in parallel with lamp (1)	2			2		2
	(b)	(i)		When current is 0.5 [A] the voltage is 0.9 ± 0.1 [V] (1) When current is 1 [A] the voltage is 2.4 ± 0.1 [V] (1) Triple would give 2.7 [V] **or** this is not triple (1) so not true To award 3 marks conclusion must be present **Alternative 1:** When the current doubles from 0.5 to 1 [A] (1) The voltage changes from 0.9 to 2.4 [V] (1) which is 2.7 times bigger **or** this is not triple (1) so not true To award 3 marks conclusion must be present **Alternative 2:** When current is 1 [A] the voltage is 2.4 ± 0.1 [V] (1) When current is 2 [A] the voltage is 8.4 ± 0.1 [V] (1) Triple would give 7.2 [V] **or** which is 3.5 times bigger **or** this is not triple (1) so not true To award 3 marks conclusion must be present **Alternative 3:** When the current doubles from 1 to 2 [A] (1) The voltage changes from 2.4 to 8.4 [V] (1) which is 3.5 times bigger **or** this is not triple (1) so not true To award 3 marks conclusion must be present			3	3		3

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				N.B. 1 When current is 0.5 [A] the voltage is 0.8 [V] (1) When current is 1 [A] the voltage is 2.4 [V] (1) This is 3 times bigger **or** this is triple (1) so true To award 3 marks conclusion must be present **N.B. 2** When voltage triples from 2 [V] to 6 [V] (1) Current changes from 0.9 [A] to 1.7 [A] (1) This is not double (1) so not true To earn credit voltages must be withing the range of 0.9 V to 8.4 V **N.B.3** A correct conclusion based on incorrect voltage readings taken from the graph award 1 mark.						
		(ii)		Voltage = 12 [V] (1) Current = 2.25 ± 0.05 [A] (1) both readings from graph Power = 27 [W] **or** 26.4 [W] **or** 27.6 [W] (1)		3		3	3	3
	(c)	(i)		Substitution: $\frac{12}{6}$ (1) = 2 [A] (1)	1	1		2	2	2
		(ii)		Straight line from origin through (12,2 **ecf**)		1		1	1	1

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		Connect in series one way **and** see if lamp lights **or** the resistance will be low **or** current will be high (1) Reverse the cell / battery / + and - / connection / box (1) See if the lamp still lights **or** the resistance will be much higher **or** diode only lets current flow one way **or** current will be zero (very small) (1) **Alternative:** Replace lamp with sealed box / add in series [with lamp] (1) Vary R and take series of reading of current and voltage (1) Reverse box / polarity of cell and repeat step 2 (1)			3	3		3
		(ii)		 Don't accept S shapes or curve with a decreasing gradient	1			1		1
				Question 1 total	**4**	**5**	**6**	**15**	**6**	**15**